

Bouba and kiki across the globe: Evidence for crossmodal correspondence in speakers of 25 different languages

Aleksandra Ćwiek^{1,2*}, Susanne Fuchs², Christoph Draxler³, Eva Liina Asu⁴, Dan Dediu⁵, Katri Hiovain⁶, Shigeto Kawahara⁷, Sofia Koutalidis⁸, Manfred Krifka^{1,2}, Pärtel Lippus⁴, Gary Lupyan⁹, Grace E. Oh¹⁰, Jing Paul¹¹, Caterina Petrone¹², Rachid Ridouane¹³, Sabine Reiter², Nathalie Schümchen¹⁴, Ádám Szalontai¹⁵, Özlem Ünal-Logacev¹⁶, Jochen Zeller¹⁷, Marcus Perlman¹⁸, Bodo Winter¹⁸

1 Humboldt-Universität zu Berlin, Institut für deutsche Sprache und Linguistik, 10099 Berlin, Germany

2 Leibniz-Zentrum Allgemeine Sprachwissenschaft (ZAS), 10117 Berlin, Germany

3 Institute of Phonetics and Speech Processing, Ludwig Maximilian University, Schellingstr. 3, 80799 Munich, Germany

4 Institute of Estonian and General Linguistics, University of Tartu, Ülikooli 18, 50090 Tartu, Estonia

5 Laboratoire Dynamique Du Langage UMR 5596, Université Lumière Lyon 2, 14 avenue Berthelot, Lyon, France

6 University of Helsinki, Department of Digital Humanities, Siltavuorenpenger 1A, 00014, Helsinki

7 Keio University, the Institute of Cultural and Linguistic Studies, 108-8345, Mita Minatoku, Tokyo, Japan

8 Bielefeld University, Faculty of Linguistics and Literary Studies, Universitätsstraße 25, 33615 Bielefeld, Germany

9 University of Wisconsin-Madison. Department of Psychology. 1202 W. Johnson St. Madison, WI 53706

10 Konkuk University, Dept. of English Language and Literature, 05029, Seoul, South Korea

11 Agnes Scott College, Asian Studies Program, 141 East College Avenue, Decatur, GA 30030, United States

12 Aix-Marseille Université, CNRS, LPL, UMR 7309, Aix-en-Provence, France

13 Laboratoire de Phonétique et Phonologie UMR 7018, CNRS & Sorbonne Nouvelle, 19 rue des Bernardins 75005, Paris, France

14 University of Southern Denmark, Department of Language and Communication, Campusvej 55, 5230 Odense M, Denmark

15 MTA Research Institute for Linguistics, Budapest, Hungary

16 Istanbul Medipol University, School of Health Sciences, Department of Speech and Language Therapy, Istanbul, Turkey

17 University of KwaZulu-Natal, School of Arts, Linguistics Discipline, Mazisi Kunene Rd., Durban 4041, South Africa

18 University of Birmingham, Dept. of English Language & Linguistics, Birmingham B15 2TT, United Kingdom

* Corresponding author: cwiek@leibniz-zas.de

Abstract (should be 150)

The highly scrutinized “bouba-kiki effect” – the widespread association of the nonce word *bouba* with a round shape and *kiki* with a spiky shape – demonstrates the potential for crossmodal correspondence between speech sounds and visual properties, which may have played a key role in language evolution. We investigated the phenomenon in a large-scale internet-based study with 897 speakers of 24 different languages from 9 different language families, followed by a field experiment with 12 technologically-remote speakers of a language from a 10th family. Overall, the results indicate strong support for the bouba-kiki effect across participants from different linguistic and cultural backgrounds, even when accounting for language genealogy and the use of a Roman writing script. Our findings show a robust crossmodal correspondence between speech sounds and visual shapes that is widely, but not universally, shared across humans who speak and write in different languages with different phonologies.

Statement of Relevance (150 words possible)

The “bouba-kiki effect”, the perceptual correspondence between the nonce word *bouba* with a rounded shape and *kiki* with a spiky shape, has been widely examined as a test bed for understanding the psychology of crossmodal correspondence between sound and vision. Key questions regard the source of the effect (e.g., word sound or orthography), and the extent to which it is a universal phenomenon. This paper reports the first large-scale controlled test of the bouba-kiki effect across participants from diverse linguistic and cultural backgrounds, which use various writing systems. Controlling for genealogy and Roman orthography, we find that the effect holds robustly among participants across the globe, evident in most, but not all, language groups, above and beyond a possible orthographic influence. Our findings provide the most extensive cross-cultural evidence to date of this highly scrutinized case of crossmodal correspondence.

Key words: perception; crossmodal association; iconicity; universals; sound symbolism

When asked to name the two shapes shown in Figure 1a using the two nonce words *bouba* and *kiki*, the majority of participants are inclined to match *bouba* with the round shape and *kiki* with the spiky one. This phenomenon – popularized by Ramachandran and Hubbard (2001), themselves inspired by Köhler’s classic work (1929) – has since been replicated numerous times in various experiments (e.g., Ahlner & Zlatev, 2010; Aveyard, 2012; D’Onofrio, 2014; Nielsen & Rendall, 2012).

Commonly known as the “bouba-kiki effect”, the mapping is generally interpreted as evidence for a crossmodal perceptual correspondence between speech sounds and visual shapes (Bremner et al., 2013; Parise & Spence, 2012; Sourav et al., 2019). Scholars have argued that such intuitive mappings between spoken form (e.g. labial consonants and vowels with lip rounding gestures) and visual meaning (e.g. rounded shape) would place “natural constraints on the ways in which sounds are mapped on to objects” (Ramachandran & Hubbard, 2001, p. 19). Thus, such mappings might have bootstrapped the formation of words in the first spoken languages (Cuskley & Kirby, 2013; Ramachandran & Hubbard, 2001).

Critical researchers have noted that the letters of the Roman alphabet used to represent “kiki” are visually spikier than the letters used to represent “bouba”, raising the possibility that orthographic shape may be a confounding factor in the effect (Cuskley et al., 2017; De Carolis et al., 2018; Hung et al., 2017; Sidhu et al., 2016). Thus, the bouba-kiki phenomenon may not reflect an actual speech-to-shape correspondence, but unimodal visual letter-to-shape matching. Consistent with this view, Cuskley et al. (2017) found that letter curvature of the written words significantly affected participants’ shape choices, even in a completely auditory task. The effect of the phonological properties of the words was argued to be secondary, with voiced consonants representing a round shape, but only in a purely auditory task. However, other experiments confirm that at least some motivation for the bouba-kiki effect stems from the sound properties of the words which cannot be attributed to orthography alone: for example, the effect holds when the stimuli are presented in either an angular or curved font (Sidhu et al., 2016).

The conclusion that sound plays an important role in the effect is further supported by cross-cultural studies. Bremner et al. (2013) demonstrated that speakers of Himba, a Bantu language spoken in Namibia, exhibit the bouba-kiki effect, even though the participants had minimal exposure to Western culture, and importantly, did not use a writing system. There is also a considerable degree of cross-cultural commonality between participants from the US and Taiwan in how they match “bouba”- and “kiki”-like sounds to shapes, despite their differing scripts (Chen et al., 2016). Some exceptions have also been found, however, suggesting that the effect is not absolutely universal. Experiments have failed to replicate the effect for Syuba speakers from the Himalayas in Nepal (Styles & Gawne, 2017), and for Songe speakers living in Papua New Guinea (Rogers & Ross, 1975). In light of these mixed results, we report here a

large-scale assessment of the strength of the bouba-kiki effect across speakers of diverse languages and writing systems.

Methods

The majority of our data was collected as part of a larger cross-linguistic survey administered online, during which participants completed a bouba-kiki task. The task occurred at the end of the roughly 15-minute survey. Participants viewed the spiky and rounded shapes shown in Figure 1a as they listened to auditory recordings of either the word *bouba* or *kiki* in two successive trials. After listening to each word, participants selected one of the two shapes that they thought better corresponded to the word (a forced choice task). The words were spoken by a female native speaker of Polish who was a trained phonetician. The order of trials was randomized without counterbalancing across participants or languages.

The survey was conducted in 24 different languages, translated by linguists or psychologists who were either native speakers of the respective languages or who arranged the translation by a native speaker. Participants were recruited by contacting native speakers of the respective languages and asking them to distribute the survey among native speakers (snowball sampling).

The cross-linguistic sample contains data from 897 participants of 24 languages from 9 different language families, shown in Table 1. Of these 24 languages, 9 do not use a Roman script: Armenian, Chinese, Farsi, Georgian, Greek, Japanese, Korean, Russian, and Thai. Figure 1b shows a map of the languages represented in our sample.

Family	Genus	Language	N of participants
Indo-European	Germanic	German	87
	Romance	Portuguese	59
	Romance	French	57
	Romance	Italian	54
	Balto-Slavic	Polish	52
	Balto-Slavic	Russian	49
	Germanic	English	41
	Hellenic	Greek	40
	Romance	Spanish	34
	Romance	Romanian	33

	Armenian	Armenian	22
	Indo-Iranian	Farsi	22
	Germanic	Swedish	21
	Germanic	Danish	18
	Albanian	Albanian	10
Finno-Ugric		Estonian	46
		Hungarian	35
Japanese (isolate)		Japanese	55
Sino-Tibetan		Chinese	48
Turkic		Turkish	38
Korean (isolate)		Korean	22
Bantu	Benue-Congo	Zulu	20
Tai-Kadai		Thai	20
Kartvelian		Georgian	14
Austronesian		Daakie	12

Table 1. Counts of participants per language and language family, ordered by number of participants within each family.

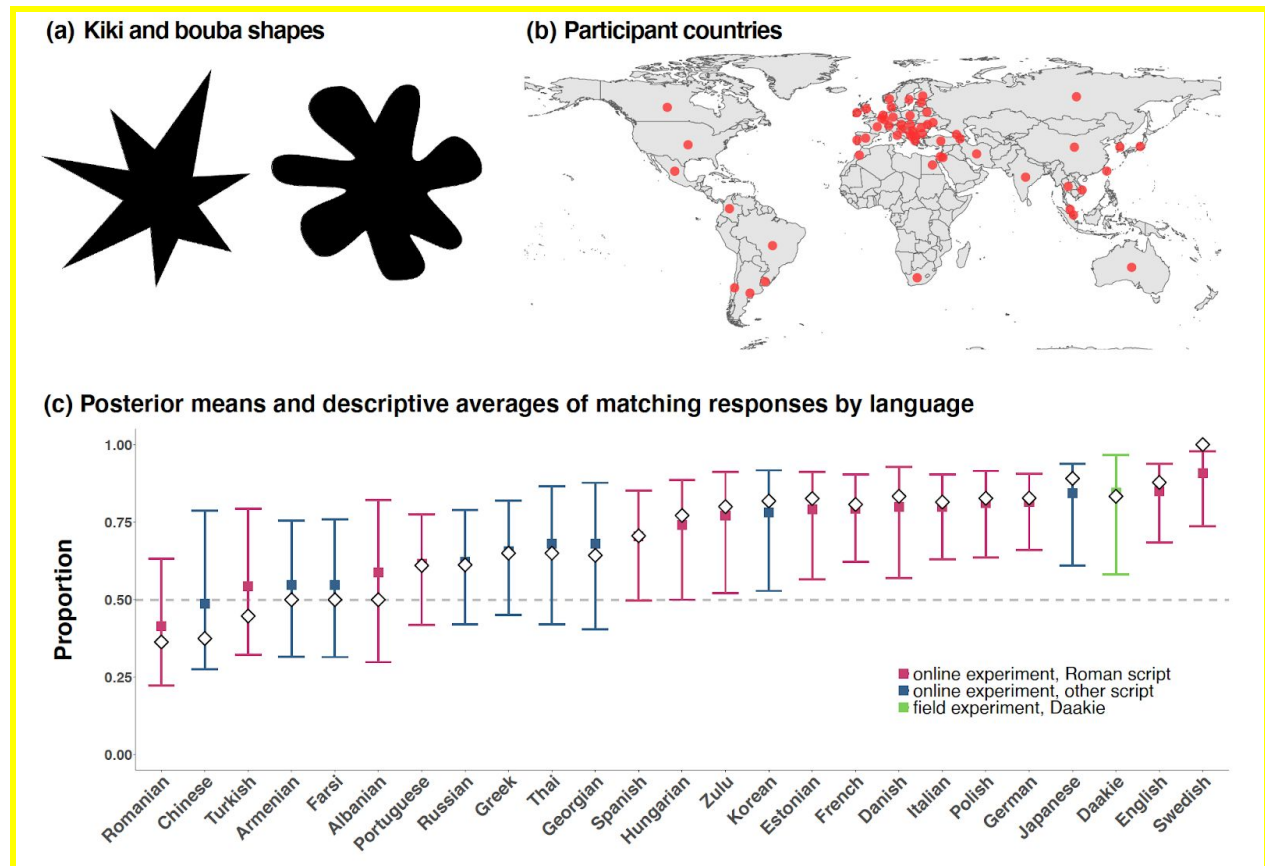


Figure 1. (a) Kiki and bouba shapes used in our experiment; (b) Map showing the countries of the participants in the online survey; (c) Posterior means (colored squares) of the proportion of matching responses (i.e., kiki for the first image in panel (a) and bouba for the second in both trials) for each individual language with the corresponding 95% credible intervals (colored vertical segments) from a Bayesian logistic regression; white diamonds indicate the descriptive averages; languages are ordered by increasing posterior means. The gray dashed line shows the baseline level.

Our analysis examined the proportion of *bouba-kiki* “matches”, which we counted only when participants matched *bouba* to the round shape and *kiki* to the spiky shape across both trials (thus treating partial matches as mismatches). To determine the overall strength of the effect across languages, we fitted a Bayesian logistic regression model, with only one fixed effects term, an intercept estimating the average proportion of participants who responded with a ‘matching’ word-to-shape alignment. This proportion was compared to a conservative baseline of $\frac{1}{2} = 50\%$, assuming complete dependence of the two trials (even though chance would be $\frac{1}{4} = 25\%$ if the two trials were independent). We assumed a normally distributed prior centered at zero ($SD = 3$) for the intercept. The random effects component of the model included random intercepts for language and language family (the isolates Japanese and Korean were treated as separate families). The model was estimated via MCMC sampling with 4 chains, each with

6,000 samples in total (of which 4,000 warm-up samples). All data and code are accessible at the following OSF repository: <https://osf.io/w7crs>.

To examine the effect of Roman orthography, we extended the above model to include a fixed effect for script (Roman versus non-Roman script, treatment coded). The random effects structure was the same as in the previous model (by-family random slopes cannot be included in this case, as only the Indo-European family contains both types of languages, i.e., those that use a Roman script and those that do not). The model was fitted with a zero-centered normally distributed prior ($SD = 3$) for the effect of script. As part of the survey, we also collected data on participants' second languages (86% of participants reported knowing more than one language), which allowed us to analyze the responses of participants without exposure to a Roman script in either their first or second languages. This group, which included just 57 participants, was not large enough to run a Bayesian model, and so for this analysis, we relied on a binomial test.

In addition to the online survey, we also conducted an experiment in the field with 12 speakers of the Austronesian (Oceanic) language Daakie. Daakie is spoken by about 1000 speakers in Port Vato, a village on the island Ambrym in Vanuatu. These participants have an agricultural lifestyle and lack regular access to electricity and to the internet. They speak an English-based creole (Bislama) with a Roman-based writing system as a second language. Children spend six years in primary school, where they are exposed to English or French. Literacy does not play a large role in their daily life. To conduct the experiment with this technologically-remote population, we designed an offline experiment that was similar to the internet-based task. Participants listened to randomized audio recordings of *bouba* and *kiki*, but, in this case, were presented with the shapes in the form of printed hard copies of the images. They made their response by pointing at the selected picture. Like the online task, this was also part of a larger multi-part experiment.

Results

The estimated posterior mean proportion of bouba-kiki matches across languages was 71%, with a 95% credible interval ranging from 57% to 82% (Figure 1c). The estimated posterior probability of the intercept being below chance level performance (50%) was very low ($< .001$). When analyzed by individual languages, the descriptive percentages of matching participants ranged from 100% (Swedish) to 36% (Romanian). Participants from 18 of the 24 languages produced bouba-kiki matches significantly more often than expected by chance. Of the other 6, Armenian, Farsi, and Albanian speakers produced matches at chance, and only Romanian, Chinese, and Turkish speakers produced matches significantly less frequently than expected by chance.

Speakers of languages with Roman scripts were 1.89 times more likely to match bouba to the round shape and kiki to the pointy shape (logit coefficient = 0.64), although matching for both groups was significantly higher than chance. The posterior mean for languages using a Roman script was 77%, compared to 64% for languages not using a Roman script. However, the evidence for the role of orthography is non-conclusive: the 95% Bayesian credible interval for the corresponding logit coefficient included zero $[-0.2, 1.52]$, with a relatively high posterior

probability of the script effect being below zero ($\beta_7 < 0 = 0.065$). We further examined the effect of the use of a Roman script in the participants' second languages, as well as their first. Of the 57 participants who did not speak a first or second language with a Roman script, we found that 38 of them (67%) produced bouba-kiki matches (significant per binomial test, $p = 0.01$).

Analysis of the field experiment with Daakie speakers found that 10 out of the 12 participants did produce bouba-kiki matches, which an intercept Bayesian logistic regression model indicates to be reliably above chance (95% Bayesian CI: 58%-97%, $\beta_0 < 0 = 0.00675$).

Discussion (313 words – count in)

We tested whether listeners from diverse linguistic and cultural backgrounds exhibited the bouba-kiki effect, matching *bouba* to a round shape and *kiki* to a spiky one. Altogether our internet survey and field experiment included 909 speakers of 25 languages from 10 language families, including languages with 10 different writing systems. Controlling for genealogical dependencies (“Galton’s problem”; Mace & Pagel, 1994), we found strong overall evidence for the effect in participants from 19 of the languages tested. Our results show that the phenomenon is reliably observed in languages with different scripts, such as Japanese, Korean, Georgian, and Thai. While speakers of languages with Roman scripts were more consistent in producing bouba-kiki matches compared to speakers of languages with different scripts, this difference was not statistically reliable. Thus, although orthography might play some role in enhancing the effect (Cuskley et al., 2017), it cannot be entirely “explained away” by orthography. This conclusion is reinforced by our results showing the effect with Daakie speakers, where along digital technology, literacy and education were relatively less widespread.

Future research remains to understand why speakers from a few languages tended not to match the common bouba-kiki alignment. Notably, some of these exceptions involved languages spoken by huge populations (Chinese), while others involved languages with Roman scripts (Turkish, Romanian). Factors could include, in the former case, a countering shape bias in the transcription of the sounds into a particular script (see Supplemental Table 1), and in both cases, the patterns could be influenced by incidental cognates with the nonce words (Turoman & Styles, 2017). On the flip side, these same factors could also serve to increase conformity for particular languages.

In summary, our experiment greatly expands the cross-linguistic study of the bouba-kiki effect, using a controlled set of stimuli with speakers from a large number of languages. The general pattern of a strong, but not universal, tendency towards the bouba-kiki effect is in line with previous cross-linguistic research (Styles & Gawne, 2017). The overall consistency of the effect suggests that it is rooted in a robust crossmodal perceptual correspondence between speech sounds and visual shapes. People from across the globe seem inclined to associate the spoken word *bouba* with a round shape and *kiki* with a spiky one, above and beyond the writing systems they might use. These kinds of widespread biases might arguably have bootstrapped

the origins and evolution of spoken languages by helping our ancestors form original connections between speech sounds and their referents (Cuskley & Kirby, 2013; Ramachandran & Hubbard, 2001).

Acknowledgments (15 words – count in)

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Author contributions

AC and SF conceived the study. AC, SF, BW, and MP designed the experiments and contributed to the writing. BW wrote the original draft and performed the statistical analysis. AC, SF, and MP gave feedback on the analysis. DD, CP, and AS performed analysis code review. CD developed the web experiment procedure. AC, SF, MP, ELA, DD, KH, SKa, SKo, PL, GO, JP, CP, RR, SR, NS, AS, ÖUL, and JZ translated (or arranged translation of) and distributed the surveys. MK carried out the field work study.

Declaration of interests

The authors declare no competing interests.

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Supplemental material

1. Online experiment

The online experiment was a part of a large-scale study on the comprehension of iconic vocalizations. After completing the main task consisting of 100 items, the participants were played a recording of spoken word *kiki* [^lkʰikʰi] or *bouba* [^lbuːba] in a randomized order. They were asked to pick a shape that depicts the recording best from among the two shapes in Figure 1a.

2. Different scripts for bouba and kiki

kiki	bouba	Roman script
кики	буба	Cyrillic script
キキ	ブーバ	Japanese
奇奇	布巴	Chinese
ㄱㅣㄱㅣ	부바	Hangul script (Korean)
คืคื	บูบา	Thai
კიკი	ბუბა	Georgian
քիքի	բութա	Armenian
κίκι	μπούμπα	Greek
کیکى	بوبا	Farsi